

Typically, PXI-based products focus on commercial industries and are based on current PC technologies. VXI products focus on military, aerospace and other high-reliability applications that require high levels of performance and demand long-term product support capabilities.

LXI is a modular standard like VXI and PXI, but does not require dedicated card cages, slot-0 interface or other proprietary link, making it cost-effective. LXI can be viewed as a distributed backplane providing the benefits of a modular card cage without the card cage, leveraging well-defined Ethernet enhancements for synchronisation and timing.

Why is VXI standard chosen over PXI in military, aerospace and other high-reliability applications?

Applications like military and aerospace are required to produce highly reliable, secure and precise test results. Because the cost of failure is typically high in these market segments, reliability and precision of measurements are critical. Longevity of the test system is another major concern because many products will be in service for 15 to 20 years and must be maintained during this period.

PC specifications change rapidly, typically every three years. These also have issues with backward-compatibility; for example, if you have a new card with PCI Express, you cannot plug it into the PCI slot or vice versa. Therefore long-term supportability is an issue.

VXI provides open hardware and open software based products. Open-architecture products are platform-, computer-, operating-system- and software-independent. Customers can confidently develop automated test equipment, using solutions from multiple vendors, without interoperability and backward-compatibility issues. VXI also provides such advantages as high performance, density and ruggedness. For this reason, the VXI platform is preferred for more powerful, high-end solutions in avionics, aerospace and defence arenas.

What kinds of tests are performed in the aerospace industry?

The defence/military and aerospace market consists of government and commercial segments involved in research, development, production and service of military and aerospace equipment and facilities. The high-technology aerospace industry is involved in the production of aircrafts, guided missiles, space vehicles, aircraft engines, propulsion units and related parts.

Data acquisition includes both electronic and mechanical tests. For example, an aircraft requires several types of testing before it is considered flight-ready. Static structural test as well as environmental test is typically performed on the airframe. Inside the aircraft, there are a large number of electronic units that must be qualified. On the whole, test applications are quite diverse and include aircraft wing-box fatigue testing, efficiency testing of jet engines, distribution

of signals during electronic testing of black boxes or line-replaceable units, and modal characterisation of launch vehicles.

Globally, which are the major countries in defence and aerospace?

The United States, France, the UK, Israel and Russia. Currently, India is not a major exporter, but it is going to become one of the largest users of defence products.

How do you see the Indian defence and aerospace market?

India is an extremely important market, and can be considered one of the golden opportunities for defence companies worldwide. In the next 10-15 years, it will replace the majority of its defence systems. This, in turn, will drive the need for automated test equipment and mechanical test systems. We aim to be a part of this.

What made you sell your products in India?

Prior to the US sanctions in India, we had strong sales in India through Microcon—the largest provider of test solutions in India. We stopped focusing on the Indian defence and aerospace market because numerous smaller companies were developing their own indigenous designs. Recently, the Indian government has allotted budget and permission for commercial equipment. Although there are companies providing modular hardware in India, these need an integrator to take modules and develop usable application configurations. Our modular products are complete application-focused instruments, with extensive built-in functionality and commercial off-the-shelf (COTS) support. In 2008, VXI Technology sold the business to a large private equity fund. Since then we have changed our model to become 'platform agnostic,' focusing not on the platform but the complete solution. In 2009, we changed our name to VTI Instruments from VXI Technology, and began concentrating on international business. Our goal for 2010 is to expand strength in India and Japan, and we are excited to provide Indian customers with a viable alternative to NI.

How do you see the future of test and measurement instrumentation?

The emergence of seamless integration of field-programmable gate arrays (FPGAs) and digital signal processor (DSP) blocks with software like MathWorks and LabVIEW allows modular instruments to be reconfigured for different applications. For example, a single data acquisition module that can be reconfigured for multiple applications. In contrast, manufacturers of the actual products (avionic units) can embed built-in test or environmental test into their products. This has led to convergence of embedded designs and test and measurement. ●